Fiscal Year:	FY 2006	**Task Last Updated:**	FY 07/27/2006
PI Name:	Gatley, Samuel J Ph.D.		
Project Title:	MicroPET Studies of Brain Damage by Heavy Ion Particles (New @ Northeastern Univ.)		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	02115	**Congressional District:**	8
Comments:	Formerly at Brookhaven National Laboratory; moved sometime in 2005.		
Project Type:	Ground	**Solicitation / Funding Source:**	02-OBPR-02
Start Date:	08/01/2005	**End Date:**	09/30/2007
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: Received NCE to 09/30/2007 per PI (7/07)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NNJ05HD38G		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	The objective of the research is to use microPET imaging as well as behavioral testing to evaluate the effects of irradiation with iron-56 particles on the rat brain. The studies were intended to help address Critical Questions 10.05 (Are there unique biological effects associate with HZE's?) and 10.07 (How can animal experiments…extrapolate the human risk…?), of the Critical Path Roadmap. The background to this grant proposal is that in studies in mice funded by the NSBRI, the PI and Dr Vazquez had shown that cocaine stimulated locomotor activity was reduced by a moderate dose (120 cGy) of 1 GeV 56Fe particles. We postulated that imaging experiments in animals might provide more sensitive and earlier indicators of damage due to HZE particles than behavioral tests. Since the small size of the mouse brain is not well suited to the spatial resolution offered by microPET, we proposed to repeat some of our studies in a rat model. We anticipated that this would enable us to identify imaging correlates of behavioral endpoints. A specific hypothesis was that changes in the metabolic rate for		

Task Description:	glucose in striatum of animals tested several months after irradiation would be correlated with alterations in locomotor activity. Other behavioral tests would be conducted as time and opportunity allowed while animals were between imaging sessions. Neurochemical effects that could be measured using PET could in principle be applied in astronauts, in terms of detecting and monitoring subtle neurological damage that might have occurred during long space missions.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Our imaging and behavioral methodologies will be relevant to evaluating effects of exposure of the brain to conventional radiation during radiation therapy of head and neck cancer. They may also be relevant to evaluation of other neurotoxic insults, such as drugs of abuse and environmental toxins such as pesticides.
Task Progress:	Rats are undergoing imaging and behavioral studies at times up to 15 months after being irradiated with 56Fe particles at NSRL at doses of 0, 120 and 240 cGy 600 MeV/n iron-56 (doses of collimated to the head region during NSRL-2, NSRL-4 and NSRL-5. Brains harvested from these animals are being evaluated using immunohistochemistry and autoradiography. Imaging: MicroPET experiments in which rats were given an i.p. injection of FDG (approximately 700 uCi), placed in a novel environment for 45 minutes during the radiotracer uptake period, anesthetized and then scanned for 20 minutes. The images were co-registered with a rat brain atlas to define brain regions. There were no significant differences across radiation dose (0, 120 or 240 cGy) or time of scanning (6 or 9 mo). Furthermore, there were no significant ($p < 0.05$) group differences for global FDG uptake. There was a very weak trend towards higher global FDG uptake in the 240 cGy group at 9 mo compared with 6 mo (21% increase; $p < 0.2$). Autoradiographic images of sections of brains from representative animals did not show obvious group differences in patterns of FDG uptake. The fact that we have not (so far) detected altered patterns of glucose metabolism in the rat brain using microPET at times up to 12 months after irradiation is superficially encouraging in that there do not appear to be large changes in brain function. This is supported by some of our behavioural data (eg open field ambulatory studies; reaction-time tests), but not by other behavioral studies (cocaine stimulated locomotion; operant task learning). Immunohistochemistry for tyrosine hydroxylase In brains harvested from NSRL-4 animals at 15 months post irradiation, we have not found any differences between 0, 120 cGy and 240 cGy radiation groups in terms of the number of tyrosine hydroxylase positive cells in the substantia nigra, or tyrosine hydroxylase staining in the striatal terminal field areas. Tyrosine hydroxylase is the key enzyme involved in synthesis of the neurotransmitter dopamine, and nigral cells containing this marker are greatly reduced in parkinsonism . This indicates that the dopamine system may be less sensitive to HZE radiation than sometimes claimed on the basis of performance in behavioral tests in which dopamine circuitry--but also other neurotransmitter systems--may be involved. Immunohistochemical studies of dopamine D1 and D2 receptors on brain sections prepared from these same animals are pending. Light-dark conditioning We have previously reported that rats exposed to 120 cGy or 240 cGy doses of iron particles learned a light-dark conditioning task more slowly than unirradiated control animals. Preliminary analysis now indicates a lack of correlation between tyrosine hydroxylase positive nigral cells and performance of individual rats. This also suggests that the dopamine system is not involved in mediating the deficits in learning. Auditory discrimination The major new studies of the last year concern auditory discrimination. Eight months after irradiation, rats were subjected to cognitive testing using an auditory conditioning protocol. Initially, responding to sound 1 (white noise) by pressing a lever was reinforced with a food pellet, but responding to sound 2 (a 2 kHz tone) was not). After learning this task, animals were presented with reversed conditions, where responding to sound 2 was reinforced but responding to sound 1 was not. In reversal learning, animals behave most adaptively by rapidly suppressing formerly reinforced behaviors, and by increasing performance of behaviors that were formerly extinguished. This study is thus taken as a test of the cognitive abilities of the animals. The number of trials necessary for animals to learn the stimulus conditions was recorded as well as a variety of other stimulus control parameters. Rats showed a radiation dose-related deficit in performance of the tasks in the absence of any other overt signs of pathology. Both groups of irradiated animals had a significant inability to discriminate the S+ (food reinforcement) and S- (extinction) stimuli when compared to controls. Both of the irradiated groups of animals were also significantly impaired in learning to decrease response rate during S- during all three phases of testing when compared to controls. Their ability to withhold responses during silence (a former S+ during pretraining) was also significantly impaired. Intertrial interval responses, which are a different way of quantifying irrelevant responses, were also significantly elevated in both irradiated groups compared to controls. Rats showed a radiation dose-related deficit in perfomance of the task in the absence of any other overt signs of pathology. There were significant effects among a variety of stimulus control parameters studied in this experiment. Both groups of irradiated animals had a significant inability to discriminate the S+ (food reinforcement) and S- (extinction) stimuli in Baseline 1 and 2 when compared to controls. Both of the irradiated groups of animals were also significantly impaired in learning to decrease response rate during S- during all three phases of testing when compared to controls. Their ability to withhold responses during silence (a former S+ during pretraining) was also significantly impaired. The ITI (intertrial interval) responses, which are a different way of quantifying irrelevant responses, was also significantly elevated in both irradiated groups compared to controls. Countermeasures

	Some of our studies are designed to evaluate Question 29d, "What are the most effective biomedical or dietary countermeasures to mitigate CNS risks?" This is being addressed by studies in which the putatively neuroprotective drug L-deprenyl (selegiline) has been administered to animals in their drinking water since irradiation. We are now collecting data from these animals. We have not seen effects of either radiation dose or deprenyl treatment on spontaneous locomotion. The results of experiments to test cocaine-stimulated locomotion are pending. Animals will shortly be euthanized and perfused with formaldehyde to allow immunohistochemical studies.
Bibliography Type:	Description: (Last Updated: 10/21/2009)
Abstracts for Journals and Proceedings	Aubele T, Kristiansen R, Rice O, Gatley SJ, Neill JC. "Long-term Effects of Whole-Brain Irradiation on Auditory Discrimination. " Paper presented at: Society for Neuroscience, 2005; Washington, D.C on 11/15/05. Society for Neuroscience abstracts, November 2005, abstract #616.19. , Nov-2005
Abstracts for Journals and Proceedings	Vargas A, Rice O, Vasquez M, Gatley S, Robinson J. "Rats Exposed to HZE Radiation Exhibit Selective Deficits in Acquisition of Light-Dark Discrimination Task. " Paper presented at: Society for Neurosciences, 2005; Washington, DC on 11/16/05. Society for Neuroscience abstracts, November 2005, abstract #883.10. , Nov-2005
Abstracts for Journals and Proceedings	Neill J, Rice O, Gatley S. "HZE irradiation impairs audition like seizures but not aging. " Paper presented at: 16th Annual NASA space radiation health investigators' workshop, 2005 on 5/16/05. 16th Annual NASA space radiation health investigators' workshop, May 2005. http://www.dsls.usra.edu/meetings/radiation2005/pdf/1045neill.pdf , May-2005
Abstracts for Journals and Proceedings	Rice O, Robinson J, Saintvictor S, Vazquez M, Gatley S. "Micropet studies of brain damage by heavy ion particles. " Paper presented at: 16th Annual NASA space radiation health investigators' workshop, 2005 on 5/15/05. Paper presented at: 16th Annual NASA space radiation health investigators' workshop, May 2005. http://www.dsls.usra.edu/meetings/radiation2005/pdf/1054rice.pdf , May-2005
Articles in Peer-reviewed Journals	Rice O, Saintvictor S, Michaelides M, Thanos P, Gatley SJ. "MicroPET investigation of chronic long-term neurotoxicity from heavy ion irradiation." AAPS Journal 2006 Aug 4;8(3):E508-14. Review. PubMed PMID: 17025269 , Aug-2006